10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 15 — Electrostatics

Solved Exercise | New Book (PECTAA 2026)

Exercise 15 (Solved)

This exercise covers electric charge, Coulomb's law, electric field and field intensity, conductors and insulators, electrostatics in daily life, electric breakdown and lightning.

A. Multiple Choice Questions

15.1 Electrical breakdown is:

(a) the process of ionizing a gas due to a strong electric field
(b) the process of neutralizing a charged object
(c) the process of charging an insulator
(d) the process of discharging a conductor

Answer: (a) the process of ionizing a gas due to a strong electric field — when the field becomes very strong the gas molecules are torn into ions and electrons, the gas becomes conducting and a spark or discharge passes through it.

15.2 Which of the following is a visible example of electrical breakdown?

(a) Corona discharge
(b) Electric current
(c) Magnetic field
(d) Electric potential

Answer: (a) Corona discharge — the faint glow (corona) around a sharp point or high-voltage wire, and the spark of lightning, are visible signs that the air has broken down.

15.3 What happens when two unlike charges are brought close to each other?

(a) They attract
(b) They repel
(c) They neutralize each other
(d) Nothing happens

Answer: (a) They attract — like charges repel and unlike charges attract, so a positive and a negative charge pull towards each other.

15.4 According to Coulomb's law, what happens to the attraction of two oppositely charged objects as their distance of separation increases?

(a) Decreases
(b) Increases
(c) Remains unchanged
(d) Cannot be determined

Answer: (a) Decreases — Coulomb's force is inversely proportional to the square of the distance, $F \propto 1/r^2$, so it becomes weaker as the charges move apart.

15.5 The device which works on the principle of electrostatics:

(a) electric bell
(b) photocopier
(c) electric heater
(d) transformer

Answer: (b) photocopier — a photocopier uses charged toner particles and a charged drum to form the image; the other three work on electromagnetism or

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heating.

B. Short Answer Questions

15.1 State Coulomb's law of electrostatics and write its mathematical form.

Coulomb's law: the electrostatic force of attraction or repulsion between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. The force acts along the line joining the two charges.

$$F \propto (q_1\, q_2) / r^2$$

$$F = k\,(q_1\, q_2) / r^2$$

Here q_1 and q_2 are the charges in coulombs, r is the distance between them in metres, and k is the constant of proportionality, called Coulomb's constant. In air or vacuum $k = 9 \times 10^9$ N m^2 C^{-2}. Force is a vector: it is repulsive for like charges and attractive for unlike charges.

15.2 What is an electric field?

An electric field is the region of space around a charge in which its electric force can be felt by another charge. If a small positive test charge q placed at a point experiences a force F, the electric field intensity at that point is E = F / q.

The field is a vector quantity; its direction is the direction of the force on a positive test charge, pointing away from a positive charge and towards a negative charge. Its unit is the newton per coulomb (N C^{-1}). An electric field is represented by field lines, drawn so that the tangent at any point gives the direction of the field and the spacing of the lines shows its strength.

15.3 Give two examples each of electrical conductors and electrical insulators.

Conductors: copper and aluminium. Both are metals with free electrons which can move through the material easily, so charge flows through them readily. Silver, iron and graphite are also good conductors, and the human body conducts electricity because of the salts dissolved in it.

Insulators: rubber and glass. In these materials the electrons are tightly bound to their own atoms, so charge cannot move through the material. Plastic, dry wood, porcelain, mica, dry air and ebonite are also insulators. That is why electric wires are made of copper but covered with rubber or plastic.

15.4 List some applications of electrostatics in daily life.

1. Photocopier and laser printer: a charged drum attracts the toner powder only where the light has fallen, so the image is transferred to the paper by electrostatic attraction.

2. Electrostatic paint spraying and powder coating: the paint droplets are charged and the object is earthed, so the paint is pulled onto it and even reaches the hidden parts. It saves paint and gives an even coat.

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3. Ink-jet printer: charged ink drops are deflected by charged plates to form letters on the paper.
4. Electrostatic precipitator: in factory chimneys the smoke particles are charged and collected on oppositely charged plates, so the smoke released into the air is clean.
5. Flue-gas and dust removal, air purifiers (ionizers) and smoke detectors work in the same way, and crop dusting aircraft charge the insecticide so that it sticks to the plants.
6. Lightning conductor, capacitors in radio and TV circuits, and the electrostatic loudspeaker all depend on electrostatics. In daily life, clothes and plastic combs pick up small bits of paper because of static charge.

15.5 What happens when like charges and unlike charges are brought close to each other?

When two like charges (both positive or both negative) are brought close to each other, they repel each other. The force acts along the line joining them and pushes them apart. When two unlike charges (one positive, one negative) are brought close, they attract each other and the force pulls them together.
In both cases the magnitude of the force is given by Coulomb's law, $F = k\, q_1\, q_2 / r^2$, so the force becomes stronger as the charges come nearer and weaker as they move apart. The rule is remembered as: like charges repel, unlike charges attract. This is the basic law of electrostatics.

15.6 What is meant by electrical breakdown, and when does it occur?

Electrical breakdown is the process in which a material that is normally an insulator — usually a gas such as air — becomes conducting because the very strong electric field in it tears the molecules into positive ions and free electrons.
It occurs when the electric field (or potential difference) across the insulator becomes larger than a certain value called the breakdown strength. For air at normal pressure this is about 3×10^6 V m^{-1}. Then the air can no longer resist the flow of charge, a sudden current passes through it and we see a spark or flash.
Examples: the spark when a charged comb is brought near a metal tap; the corona discharge round a pointed high-voltage wire; and lightning, where the huge field between a cloud and the ground breaks down a path of air and a massive current flows. Breakdown is prevented by insulating high-voltage cables thickly and by rounding off the sharp edges of conductors.

15.7 What are the dangers of static electricity?

1. Sparks and shocks: the charge built up on a person walking on a synthetic carpet can give a painful shock when a metal door handle is touched, and it can make a person drop a tool.
2. Fire and explosion: a single spark near petrol, diesel, LPG, or in a flour mill, oil tanker or grain silo can ignite the fuel vapour or the dust cloud and cause a serious explosion. This is why fuel tankers are earthed and chain-dragged before loading.

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3. Damage to electronic equipment: the static charge on a person's body can destroy chips, transistors and hard disks when they are handled; computer repairers wear an earthed wrist strap for this reason.
4. Dust attraction: charged TV screens and monitors attract dust, and charged plastic surfaces attract dust and lint, which spoils photographic film and precision parts.
5. Nuisance in industry: static charge makes plastic sheets and fibres stick together or to machinery and causes jams in printing and textile mills. These dangers are reduced by earthing, using antistatic sprays and humidifying the air.

C. Constructed Response Questions

15.1 Why do some materials gain electrons while others lose electrons when rubbed together? What property of the materials determines this behaviour?

When two different materials are rubbed together, the atoms at the surfaces come very close and some electrons are pulled from one surface to the other. The material whose atoms hold their electrons only weakly loses electrons and becomes positively charged, while the material whose atoms hold them more strongly gains electrons and becomes negatively charged. The charge is never created; it is only transferred, which is why the two bodies get equal and opposite charges.

The property that decides the direction of transfer is called the electron affinity (or work function) of the material — how strongly the atoms of that material hold on to their outer electrons. Each material can be placed in a series, the triboelectric series, in which glass, wool and fur tend to lose electrons while rubber, plastic and ebonite tend to gain them. That is why a glass rod rubbed with silk becomes positive, and an ebonite or plastic rod rubbed with fur becomes negative.

The amount of charge that can be built up also depends on the area of contact, the dryness of the air (moist air leaks the charge away) and the insulating property of the materials, which does not allow the transferred electrons to flow back.

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15.2 A charged comb attracts small pieces of paper, but after some time, the paper pieces fall off. Why does this happen?

When the charged comb is brought near a small piece of paper, the charge on the comb repels the like charges inside the paper and attracts the opposite charges towards the near side. The near side of the paper therefore becomes oppositely charged, and because it is nearer to the comb than the far side, the attractive force is stronger than the repulsive force — so the paper is attracted and jumps to the comb.

While the paper hangs on the comb, the charge from the comb slowly leaks through the surface of the paper and the moist air, and the paper also picks up some of the comb's charge. Then the paper and the comb carry the same kind of charge, and they repel each other. The paper is pushed away and falls off under its own weight.

The same thing is seen with a rubbed balloon sticking to a wall and then dropping after some time: the charge leaks away to the damp air and the wall, and the

attraction disappears.

15.3 Why do dust particles often get attracted to television screens or computer monitors after they are turned ON?

When a television screen or a computer monitor is switched on, a high potential difference is applied across its tube or panel. Electrons are emitted from the cathode (or the discharge takes place in the panel), and the front surface of the screen gradually collects a large electric charge. So the screen becomes electrostatically charged.

The fine dust particles floating in the air are neutral, but when they come near the charged screen, their charges separate by induction: the opposite charge is drawn to the side facing the screen. The result is an attractive force, and the dust particles stick to the glass.

This is why the screen becomes dusty soon after use and why we see a layer of dust when we wipe it. The dust is held so long as the charge remains; when the set is switched off and the charge leaks away, the dust falls off or can be wiped easily. The same effect makes the screen crackle faintly and attract hair when touched after use.

15.4 If we rub a balloon on our dry hair and place it near a thin stream of water, the water bends toward the balloon. What does this tell us about the nature of water molecules?

The rubbed balloon becomes charged (negatively, because rubber gains electrons from hair). When it is brought near the thin stream of water, the stream is pulled towards the balloon, so a force of attraction acts between them.

A neutral stream would not be attracted unless its charge distribution could be rearranged. This shows that the water molecules are polar — each molecule has a partial positive charge on its hydrogen side and a partial negative charge on its oxygen side, so the whole molecule, though neutral overall, behaves like a tiny dipole. In the electric field of the balloon these molecules turn and line up, with the opposite ends towards the balloon, and are then attracted by it.

So the experiment tells us that water molecules are polar (dipoles) and that they can be polarised and attracted by a charged body. The same property, together with the hydrogen bond between water molecules, explains why water is such a good solvent and why a charged comb can deflect a fine stream from a tap.

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15.5 A conductor and an insulator are both placed in an electric field. How will their behaviour differ, and why?

In a conductor there are free electrons which can move easily through the material. When it is placed in an electric field, these electrons drift in a direction opposite to the field until they reach the surface. The surface of the conductor thus acquires a charge, which produces its own field inside the conductor that exactly cancels the applied field. As a result, in the steady state the electric field inside a conductor is zero, and the charge resides only on its outer surface. The conductor is said to be polarised (or its charge redistributed), and it can shield a region from outside fields — this is electrostatic shielding.

In an insulator the electrons are bound to their atoms and cannot move through the material. When it is placed in a field, only a slight displacement of the electron cloud of each molecule takes place, so each molecule becomes a tiny dipole aligned with the field. The insulator is said to be polarised and acquires a small induced charge on its surfaces, but no free charge flows and the field inside is not reduced to zero; it is only reduced a little. No current can pass through it.

So the difference is one of degree of freedom of electrons: in a conductor charge moves freely and the internal field becomes zero, while in an insulator the charge is only slightly displaced and no charge flows. That is why metals are used for cables and rubber or plastic for insulation.

D. Comprehensive Questions

15.1 Discuss the working of a gold-leaf electroscope. How can it be used to determine the nature and magnitude of charge on a body?

Construction. An electroscope consists of a metal rod fixed through an insulating stopper in a glass jar. At the top of the rod is a metal disc or knob, and at the lower end inside the jar two thin leaves of gold (or aluminium) are attached. The jar protects the leaves from air currents and moisture.

Working. When a charged body is brought near or touched to the knob, the same kind of charge spreads down the rod to the leaves. Both leaves get the same charge, so they repel each other and diverge. The greater the charge, the greater the repulsion and the larger the angle between the leaves. When the charge leaks away, the leaves collapse again.

To find the nature of the charge. First charge the electroscope with a known charge — say, by touching it with a positively charged glass rod; the leaves diverge. Now bring the unknown charged body near the knob without touching. If the leaves diverge further, the unknown charge is of the same kind (positive); if the leaves come closer together, it is of the opposite kind (negative). This is called the method of testing by induction. Alternatively, touch the body to the knob and then test with a known rod.

To compare the magnitude of charges. Charge the electroscope with a small charge and note the divergence of the leaves. Then bring a body of the same kind but unknown amount of charge and touch it to the knob; the divergence increases. Comparing this divergence with the divergence produced by charges of known

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amount gives a measure of the charge. So the angle between the leaves tells us the relative magnitude of the charge, and the direction of the change in divergence tells us its nature.

15.2 How does Coulomb's law help in understanding the interaction between charged particles? What factors affect the force between them?

Coulomb's law gives the exact size and direction of the electrostatic force between two point charges: $F = k\, q_1\, q_2 / r^2$. With it we can understand why a rubbed balloon sticks to a wall, why the leaves of an electroscope repel, why an electron stays in its orbit around the nucleus, and why two charged spheres push or pull each other with a force we can calculate in advance. It also explains why the electrostatic force becomes enormous when charges are brought very close, and why it falls off very quickly as they are separated.

Factors affecting the force:

(i) Magnitude of the charges — the force is directly proportional to the product $q_1 q_2$; doubling either charge doubles the force.

(ii) Distance between them — the force is inversely proportional to r^2; doubling the distance reduces the force to one quarter.

(iii) The medium between the charges — in a medium of relative permittivity ε_r the force becomes $F = k\, q_1 q_2 / (\varepsilon_r\, r^2)$, so it is weaker in water ($\varepsilon_r = 80$) than in air.

(iv) The nature of the charges — like charges repel and unlike charges attract, so the direction of the force depends on their signs. The force also acts along the line joining the charges and obeys Newton's third law: each charge experiences an equal and opposite force.

15.3 Explain the concept of electric field intensity. How can the electric field of a point charge and a parallel plate capacitor be represented using field lines?

Electric field intensity at a point is the electrostatic force experienced per unit positive charge placed at that point: $E = F / q$. It is a vector directed along the force on a positive charge, its unit is N C^{-1} (or V m^{-1}), and for a point charge q at distance r it has the value $E = k\, q / r^2$.

Field lines are imaginary lines drawn in an electric field such that the tangent at any point gives the direction of the field there. Their rules: they start on positive charge and end on negative charge; they never cross one another; they are closer together where the field is strong and farther apart where it is weak; and they leave or enter a conductor at right angles to its surface.

Field of a point charge. For an isolated positive charge the lines are straight and point radially outward, becoming farther apart as we go away — showing that the field becomes weaker. For a negative charge the lines point radially inward. For two equal and opposite charges (a dipole) the lines curve from the positive to the negative charge.

Field of a parallel plate capacitor. Between two large parallel plates, one positive and one negative, the field lines are straight, parallel and equally spaced, running from the positive to the negative plate. This shows that the field between the plates

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is uniform — the same in magnitude and direction at every point. Near the edges the lines bulge outward (fringing). The field strength is E = V / d, where V is the potential difference between the plates and d is their separation.

15.4 Explain how and why electrical conductors and insulators behave differently in terms of charge transfer and movement of electrons.

Conductors are materials in which the outermost electrons of the atoms are loosely bound. In a metal these electrons form a 'sea' of free electrons that can move throughout the material. When a conductor is given charge, or connected to a source of potential difference, these free electrons drift through the lattice and carry charge from one place to another. Charge can therefore be transferred easily, and it spreads at once over the outer surface of the conductor, making the field inside zero.

Insulators have their electrons tightly bound to individual atoms or molecules, with no free electrons. Charge placed on an insulator cannot move through it; it stays where it was put, on the small area that was rubbed. The only movement possible is a slight displacement of the electron cloud of each molecule (polarisation), which produces a small induced surface charge but no flow of charge. So an insulator does not conduct.

Why the difference? It comes from the atomic structure and the strength of the force holding the electrons to their atoms. In metals the outer electrons are almost free (low ionisation energy), while in insulators they are bound in covalent or ionic bonds and need a very strong field — enough to cause breakdown — to be pulled out.

Practical consequences: wires are copper or aluminium so that current flows easily, and they are covered with rubber, PVC or porcelain so that the charge cannot leak out or reach the user; capacitors use a conducting plate pair separated by an insulating dielectric; and earthing a conductor removes its charge at once, which is impossible for an insulator without a conducting path.

15.5 Discuss how electrostatics plays a role in real-life hazards, such as dust explosions in factories and static shocks from synthetic clothing. How can these hazards be minimized?

Dust explosions in factories. In flour mills, sugar refineries, coal mines and grain silos fine dust floats in the air. Friction of the moving dust particles, or of belts and machinery, builds up static charge on them. A single spark — from static discharge, a machine or a person — can ignite the dust cloud, and the flame spreads through the whole building with tremendous force. Similar danger exists where petrol vapour, LPG or solvent vapour is present, as in oil tankers and paint shops.

Static shocks from synthetic clothing. Synthetic fibres such as polyester, nylon and acrylic do not conduct charge, so when they rub against the body or against each other a large static charge builds up. When the wearer touches a metal object, or gets out of a car, this charge jumps as a spark — the familiar small shock. In a hospital or an electronics workshop such a spark can be dangerous, and for a person with a heart pacemaker it can be serious.

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Other hazards: sparks near flammable gases at petrol stations, damage to computer chips and electronic instruments, jams and fire in textile and printing machines, and dust on films and photographic material.

How to minimize the hazards: (i) earth (ground) all machinery, tanks and tankers so that the charge flows away at once — fuel tankers drag a metal chain; (ii) use antistatic sprays, additives in fuels and conducting floor tiles and footwear in industry; (iii) keep the air moist, because damp air carries the charge away; (iv) wear cotton clothes and use antistatic wrist straps when handling electronic components; (v) provide proper ventilation to remove dust and vapour, and fit lightning conductors and flameproof electrical equipment; (vi) touch a metal object with a key or a coin before touching it with the hand, to let the charge discharge slowly and avoid a painful spark.

E. Numerical Problems

Use $k = 9 \times 10^9$ N m^2 C^{-2}, and quote charges in coulombs (1 μC = 10^{-6} C).

15.1 A positive test charge of 25 μC is placed in an electric field. The force on it is 0.500 N. What is the magnitude of the electric field at the location of the test charge?

Given: $q = 25\ \mu C = 25 \times 10^{-6}$ C
$F = 0.500$ N

To find: $E = ?$

Formula: $E = F / q$

Solution:
$E = 0.500 / (25 \times 10^{-6})$
$E = 2.0 \times 10^4$ N C^{-1}

Answer: $E = 2.0 \times 10^4$ N C^{-1}

15.2 Two point charges, q_1 = 8 μC and q_2 = −4 μC, are placed at a distance of 120 cm. What will be the Coulomb's force between them? Also, find the nature of the force.

Given:
$q_1 = 8 \times 10^{-6}$ C
$q_2 = 4 \times 10^{-6}$ C (magnitude)
$r = 120$ cm $= 1.2$ m
$k = 9 \times 10^9$ N m^2 C^{-2}

To find: $F = ?$, nature of force = ?

Formula: $F = k\, q_1\, q_2 / r^2$

Solution:
$F = (9 \times 10^9 \times 8 \times 10^{-6} \times 4 \times 10^{-6}) / (1.2)^2$
$F = (9 \times 10^9 \times 32 \times 10^{-12}) / 1.44$
$F = 0.288 / 1.44$
$F = 0.2$ N
The charges are of opposite signs, so the force is attractive.

Answer: F = 0.2 N, attractive (the book writes 'direction of repulsion'; the charges are unlike, so the force between them is attractive)

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15.3 Two identical charges repel each other with a force of 0.2 N when they are 9 cm apart. Find the force between them when they are 3 cm apart.

Given: $F_1 = 0.2$ N
$r_1 = 9$ cm
$r_2 = 3$ cm
charges and hence $q_1 q_2$ are unchanged

To find: $F_2 = ?$

Formula: $F \propto 1 / r^2 \Rightarrow F_2 = F_1 (r_1 / r_2)^2$

Solution: $F_2 = 0.2 \times (9 / 3)^2$
$F_2 = 0.2 \times 9$
$F_2 = 1.8$ N

Answer: F = 1.8 N

15.4 A point charge q = 10 μC is placed in air. Calculate the electric field strength at a distance of 0.5 metres from the charge.

Given: $q = 10\ \mu C = 10 \times 10^{-6}$ C
$r = 0.5$ m
$k = 9 \times 10^9$ N m^2 C^{-2}

To find: $E = ?$

Formula: $E = k\, q / r^2$

Solution: $E = (9 \times 10^9 \times 10 \times 10^{-6}) / (0.5)^2$
$E = (9 \times 10^4) / 0.25$
$E = 3.6 \times 10^5$ N C^{-1}

Answer: $E = 3.6 \times 10^5$ N C^{-1}

15.5 Two small equally charged metal spheres having charge 3 μC are brought close together such that the distance between them is 2.0 cm. Calculate the magnitude of repulsive force that each sphere exerts on the other.

Given: $q_1 = q_2 = 3 \times 10^{-6}$ C
$r = 2.0$ cm $= 0.02$ m
$k = 9 \times 10^9$ N m^2 C^{-2}

To find: $F = ?$

Formula: $F = k\, q_1\, q_2 / r^2$

Solution: $F = (9 \times 10^9 \times 3 \times 10^{-6} \times 3 \times 10^{-6}) / (0.02)^2$
$F = (9 \times 10^9 \times 9 \times 10^{-12}) / 4 \times 10^{-4}$
$F = (8.1 \times 10^{-2}) / (4 \times 10^{-4})$
$F = 202.5$ N

Answer: F = 202.5 N

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15.6 Two small equally charged metal spheres repel each other with a force of 0.5 N when placed 3 cm apart in the air. Calculate the charge on each sphere.

Given: $F = 0.5$ N
$r = 3$ cm $= 0.03$ m
$q_1 = q_2 = q$
$k = 9 \times 10^9$ N m^2 C^{-2}

To find: $q = ?$

Formula: $F = k\,q^2 / r^2 \Rightarrow q = \sqrt{(F\,r^2 / k)}$

Solution:
$q^2 = (0.5 \times (0.03)^2) / (9 \times 10^9)$
$q^2 = (0.5 \times 9 \times 10^{-4}) / (9 \times 10^9)$
$q^2 = 5 \times 10^{-14}$
$q = 2.24 \times 10^{-7}$ C
$q = 0.224$ μC

Answer: $q = 2.24 \times 10^{-7}$ C ≈ 0.224 μC on each sphere

15.7 A force of 85 N acts between two charges placed 3.2 cm apart. If one charge is 25 μC, determine the other charge.

Given: $F = 85$ N
$r = 3.2$ cm $= 0.032$ m
$q_1 = 25$ μC $= 25 \times 10^{-6}$ C
$k = 9 \times 10^9$ N m^2 C^{-2}

To find: $q_2 = ?$

Formula: $F = k\,q_1\,q_2 / r^2 \Rightarrow q_2 = F\,r^2 / (k\,q_1)$

Solution:
$q_2 = (85 \times (0.032)^2) / (9 \times 10^9 \times 25 \times 10^{-6})$
$q_2 = (85 \times 1.024 \times 10^{-3}) / (2.25 \times 10^5)$
$q_2 = 8.704 \times 10^{-2} / 2.25 \times 10^5$
$q_2 = 3.87 \times 10^{-7}$ C

Answer: $q_2 = 3.87 \times 10^{-7}$ C (about 0.39 μC)

The book prints 3.87 μC for this answer. Since 3.87×10^{-7} C $= 0.387$ μC, the charge is 3.87×10^{-7} C; in the paper show the working and then write the answer in scientific notation.

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Formulae to Remember

- Coulomb's law: $F = k\,q_1\,q_2 / r^2$, $k = 9 \times 10^9$ N m^2 C^{-2}
- Electric field: $E = F / q$, and for a point charge $E = k\,q / r^2$
- Field between parallel plates: $E = V / d$
- 1 μC $= 10^{-6}$ C; like charges repel, unlike charges attract